ABDULBASIT AJIBOWU

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EDUCATION

Bachelor of Science (B.Sc.) in Electronic and Computer Engineering
Lagos State University | 2025

Key Coursework: Embedded Systems & Firmware Engineering, Digital Logic Design,Data Structures & Algorithms, Database Management Systems (DBMS), Microprocessor & Microcontroller Systems, Digital Logic Design, Control Systems Engineering, Computer Networks, and Operating Systems.

TECHNICAL & ENGINEERING SKILLS

- Programming & Scripting: C++, Python, JavaScript, and SQL
- Hardware & Embedded Engineering: Arduino, AVR Microcontrollers, Firmware Development, Sensors & Actuators, GSM/Bluetooth Modules, Motor Drivers, Breadboarding & Circuit Prototyping
- Backend & Systems Development: Node.js/Express, RESTful APIs, Relational Databases (PostgreSQL, MySQL), System Architecture, Object-Oriented Programming (OOP)
- Digital & Analog Electronics: Transistor-Level Logic Design, Circuit Design, Sensor Feedback Loops, Low-Level Serial Protocols (UART, SPI, I2C)
- Tools & Platforms: Git/GitHub, Postman, Kicad, Fusion360, Proteus (Circuit Simulation), Arduino IDE and Visual Studio Code Editor.

ACADEMIC & ENGINEERING PROJECTS

Automated Shoe Polishing Machine

- Designed and built a multi-actuator mechatronic platform utilizing microcontrollers, analog sensor feedback loops, and real-time control logic.
- Developed embedded firmware in C++ to coordinate sequential motor actuation, timing sequences, and dynamic speed controls based on sensor inputs.
- Implemented feedback control loops to regulate mechanical force and ensure precise automated operation without surface damage.

Electronic ID Card System Design | Embedded Systems & Smart Security

- Co-designed and developed a prototype electronic ID card system integrating embedded microcontrollers, RFID/NFC modules, and secure data storage for identity verification.
- Engineered low-power circuit schematics and hardware architecture to optimize power consumption for portable, handheld operation.
- Programmed embedded firmware to manage encrypted data transmission between the smart ID card and authentication readers, ensuring secure access control.

Smart Home IoT Monitoring System

- Engineered an automated home monitoring and control framework using an Arduino microcontroller paired with an HC-05 Bluetooth module for real-time mobile interaction.
- Configured low-level serial communication protocols (UART) to execute wireless commands from a mobile device to actuate connected home appliances, including light bulb sockets.

Bluetooth-Controlled 4-Wheel Robotic Car

- Designed and constructed a four-wheel mobile robotic platform utilizing an Arduino controller, dual H-Bridge motor drivers, and a Bluetooth serial interface (HC-05).
- Wrote custom firmware to process incoming serial data streams and execute real-time differential steering and velocity controls.
- Optimized power distribution architecture to isolate motor noise from sensitive logic circuitry, preventing control system resets.

Sound Detector Safety Alarm System

- Engineered a real-time, event-triggered safety system utilizing high-sensitivity acoustic sensors, microcontrollers, and visual/auditory output actuators.
- Designed hardware threshold calibration and software debouncing routines to minimize false triggers from ambient acoustic noise.
- Integrated interrupt-driven execution routines to ensure sub-millisecond response latency upon threshold activation.

Transistor-Level Logic Gate Modeling

- Designed and verified fundamental digital logic gates (AND, OR, NOT, NAND, NOR) constructed strictly from discrete bipolar junction transistors (BJTs) and MOSFETs.

- Analyzed physical logic behavior, noise margins, switching delays, and signal performance characteristics at the semiconductor level.
- Validated experimental voltage outputs against theoretical Boolean logic tables using multimeter and oscilloscope diagnostics.

Learning Management System (LMS) | Backend (Software Engineering)

- Architected a scalable backend service featuring full CRUD functionality to manage user interactions and digital learning resource distribution.
- Implemented Role-Based Access Control (RBAC) across administrator, instructor, and student tiers, ensuring fine-grained access security.
- Designed normalized relational database schemas in MySQL to optimize query performance and ensure data integrity.

AWARDS & INNOVATION COMPETITIONS

AgroInnovatex Challenge - 1st Place Winner | AQUATECH Project

Designed AQUATECH, an award-winning automated aquaculture management system designed to bridge technology and precision agriculture with microprocessors, environmental sensors, and mechanical actuators to automate real-time feeding cycles and water quality/exchange management.

ADVANCED TRAINING & CERTIFICATIONS

Microsoft Certified: Power Platform Fundamentals (PL-900)
June 2026

Junior Sharepoint Developer (Microsoft Sharepoint)
Exec200 Systems Limited
April 2026 - till present

Software Developer Intern
Exec200 Systems Limited
July 2024 – April 2026

Software Development Certification - Power Learn Project Africa
December 2024
Focus: Python, Web Development, Database Management, Dart With Flutter, REST API Architecture, Security Protocols (JWT, OAuth), Microservices Fundamentals.

Laboratory Assistant
Students Work Experience Program (SWEP)
Lagos State University, Electronic and Computer Engineering Laboratory
December 2023 – February 2024

Machine Operator - Adlan Engineering Company
Ejigbo Lagos State
August 2017 – December 2017

- Fabricated and refabricated complex machine parts using lathes and milling machines.
- Executed precision welding, grinding, and gear machining operations to support equipment maintenance and production.

LEADERSHIP & VOLUNTEER EXPERIENCE

Executive Secretary
Muslim Students' Society of Nigeria (MSSN LASU)
August 2024 – July 2025

Honourable Member
25TH Engineering Society Parliamentary Council (ESPC) Assembly